ADVERTISING SALES PREDICTION USING MACHINE LEARNING

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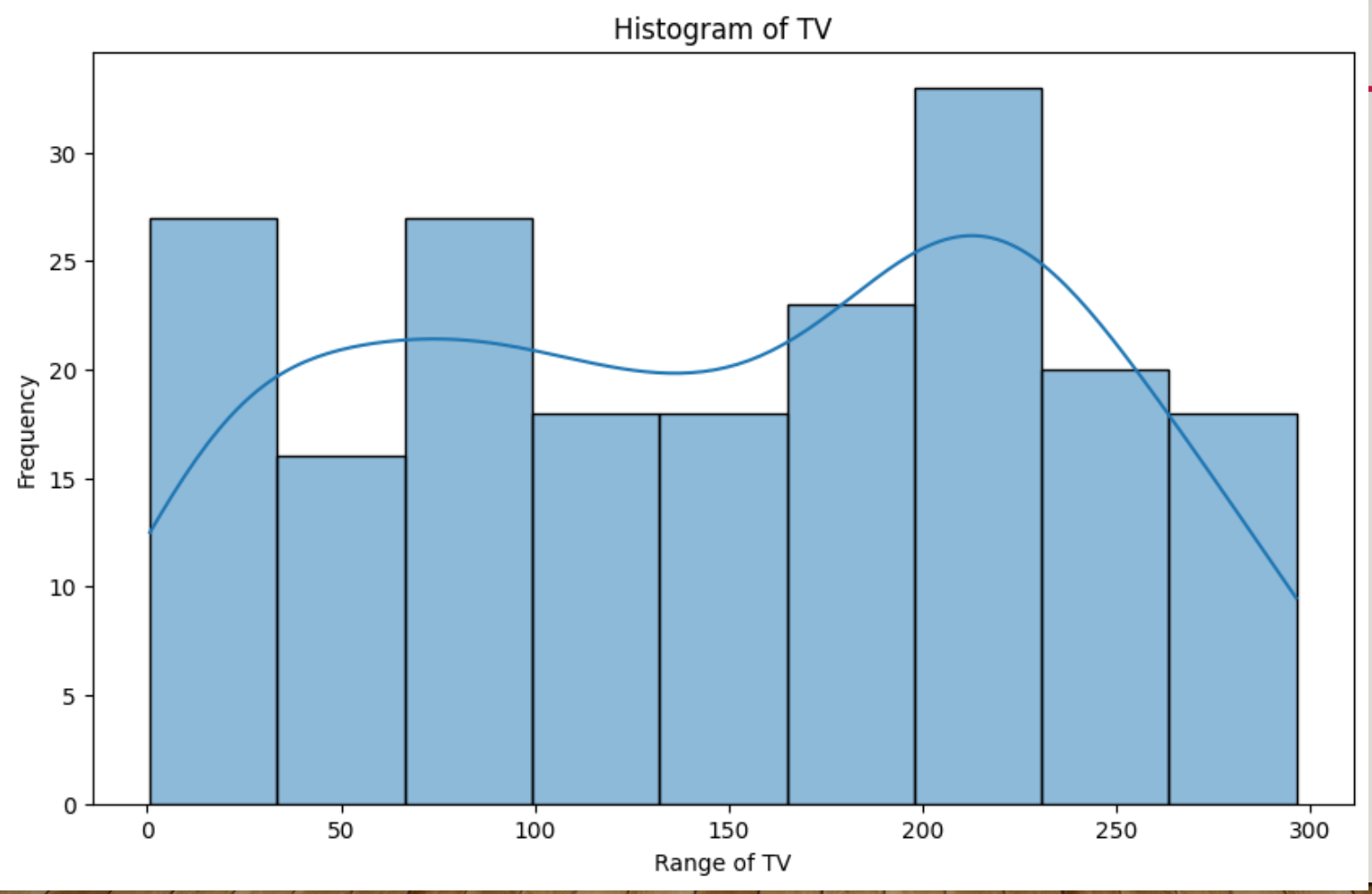
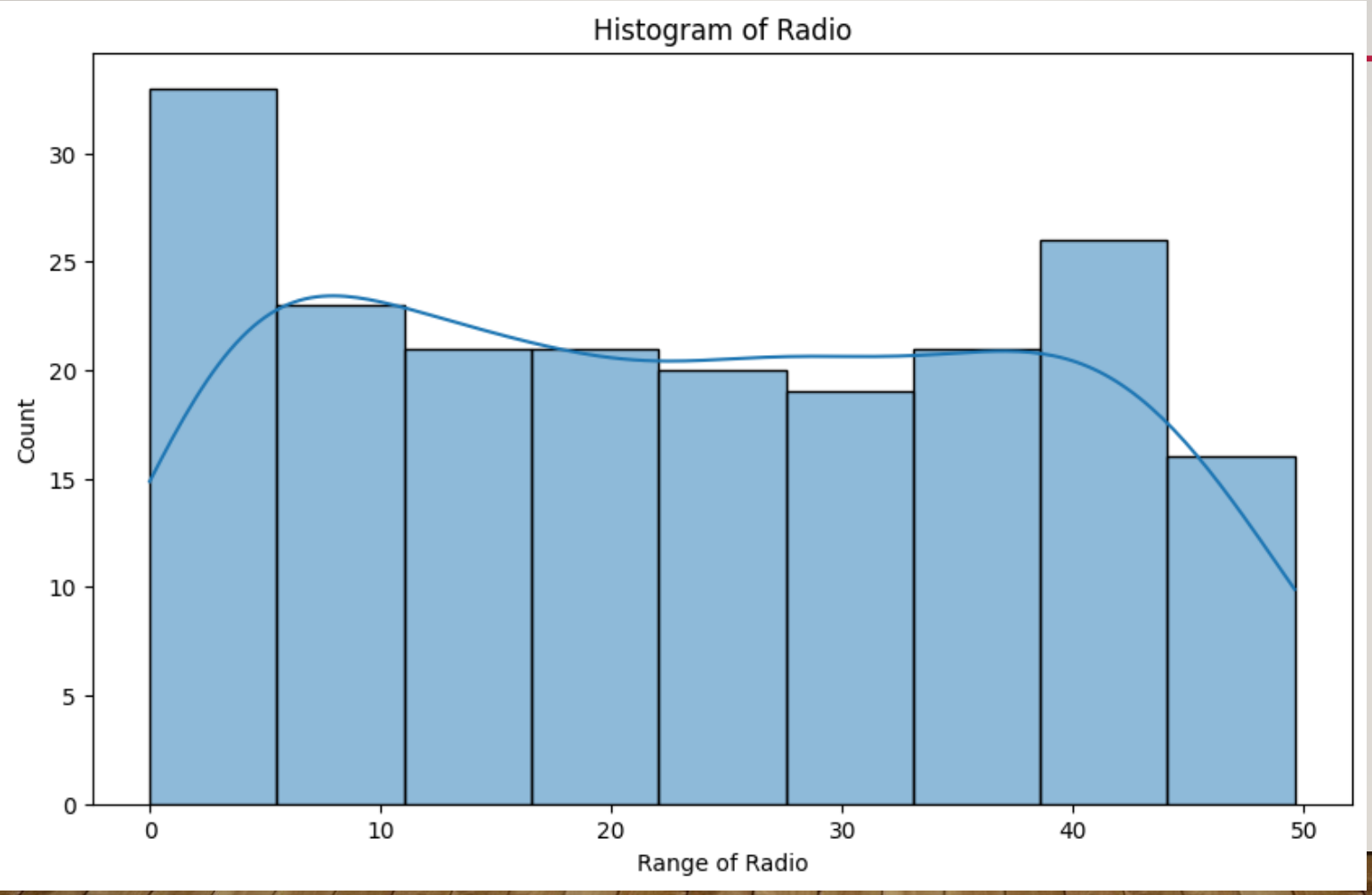
Data science

HISTOGRAM OF TV



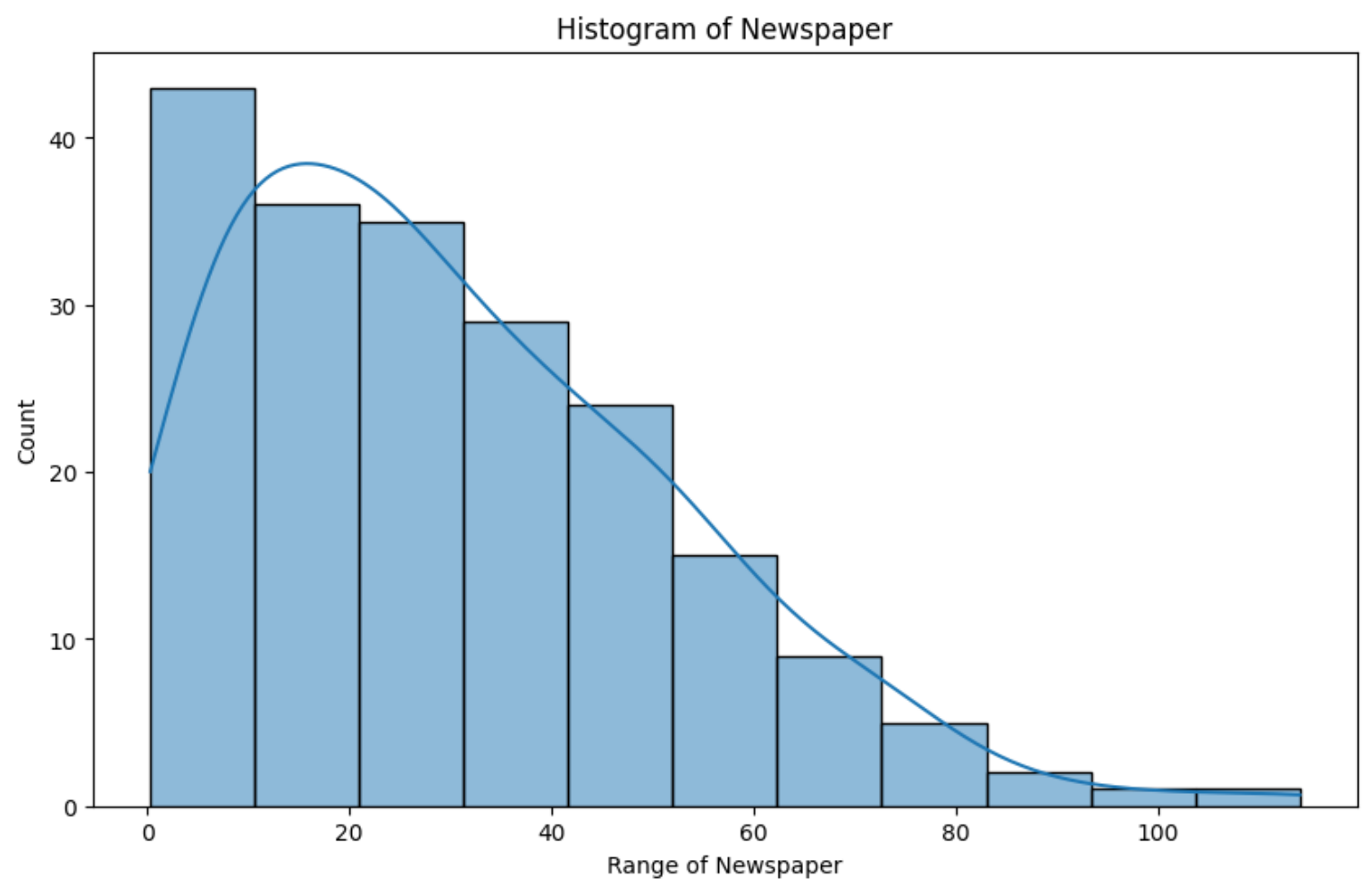
Uniform graph and curve

HISTOGRAM OF RADIO



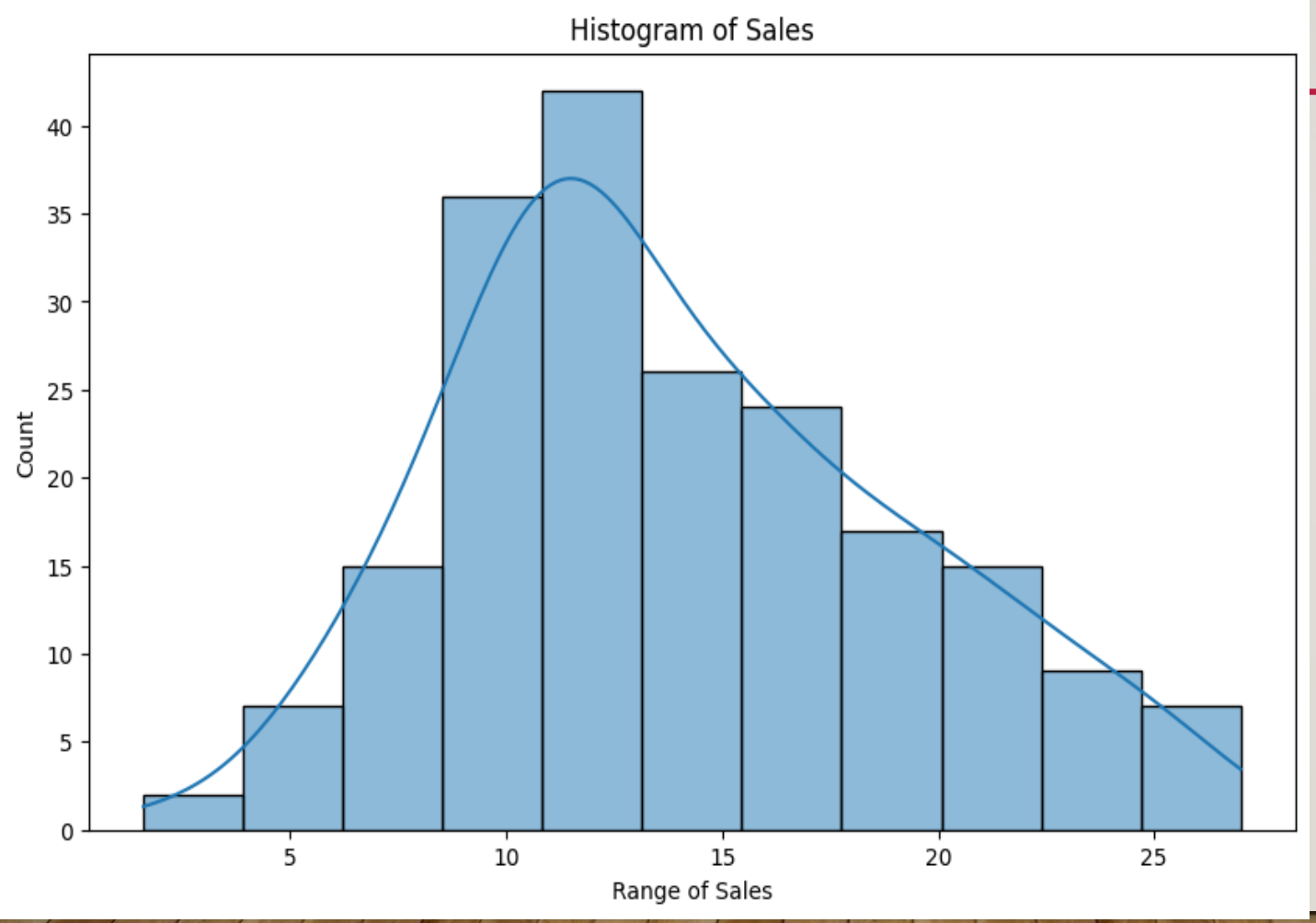
Uniform graph and curve

HISTOGRAM OF NEWSPAPER



Skew curve and gra

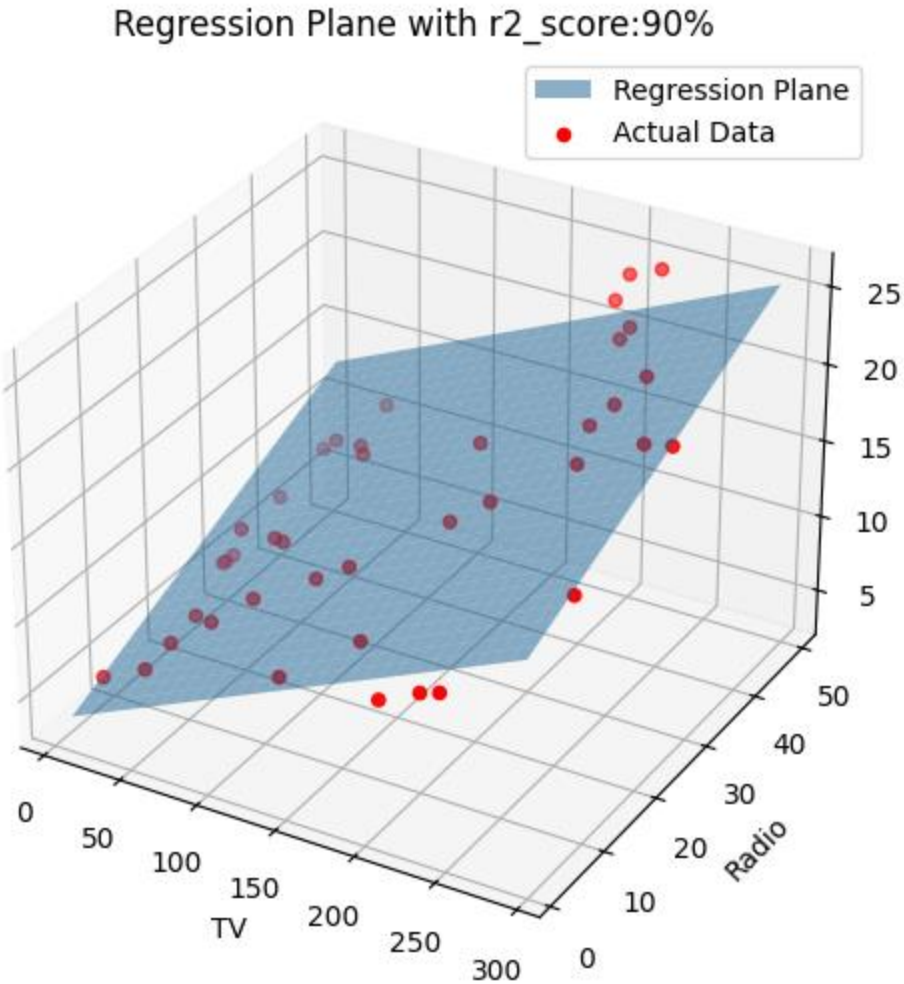
HISTOGRAM OF SALES

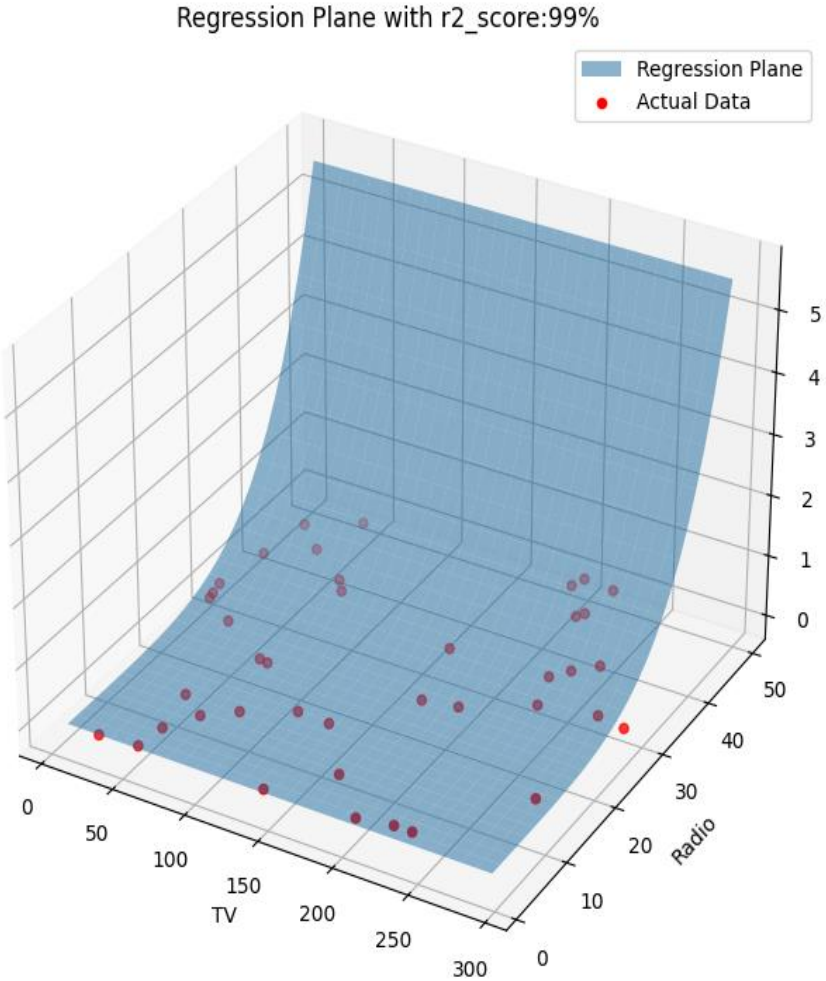


Bell shape graph and curve

CONCLUSION

- · Histograms help visualize how advertising budgets and sales are distributed.
- • TV advertising shows higher spending compared to other media channels.
- • Radio advertising has a relatively balanced distribution.
- • Newspaper advertising is skewed toward lower spending values.
- • Sales distribution indicates most products achieve moderate sales levels.
- • These visualizations help understand the dataset before applying machine learning models such as Linear Regression.


Regression Plane with r2_score:90%
Regression Plane
Actual Data
TV
Radio
0
50
100
150
200
250
300
0
10
20
30
40
50
5
10
15
20
25


Regression Plane with r2_score:99%
Regression Plane
Actual Data
TV
Radio
0
50
100
150
200
250
300
0
10
20
30
40
50
0
1
2
3
4
5

CONCLUSION:

- Linear only has 90% r2_score
- Polynomial has 99% r2_score
- So polynomial model explains better the data